WIDE-BANDGAP MATERIALS

Diamond electronics with high carrier mobilities

Field-effect transistors based on heterojunctions of hydrogen-terminated diamond and hexagonal boron nitride can offer surface carrier mobilities as high as 680 $cm^2\ V^{-1}\ s^{-1}$.

Moshe Tordjman

The development of high-power and high-frequency electronics with improved performance is needed to meet the ever-increasing energy demands of the electrical power industry. Devices based on wide-bandgap materials were traditionally seen as best suited for this task because they can sustain fast switching frequencies and have high power densities and a decent thermal dissipation rate. Nevertheless, traditional wide-bandgap materials that are currently in wide use, such as gallium arsenide (GaAs), gallium nitride (GaN) and silicon carbide (SiC), show limitations in their performances due to poor physical material characteristics.

A promising alternative wide-bandgap material is diamond. It offers a high critical electric breakdown field (more than 10 MV cm^{-1}), thermal conductivity (more than 20 W $cm^{-1}\ K^{-1}$) and intrinsic carrier mobility (more than 3,500 $cm^2\ V^{-1}\ s^{-1}$), and is expected to outperform GaAs, GaN and SiC (ref. [1]). But realizing the full potential of diamond-based electronic devices has proved challenging due to the difficulties involved in transforming the material from an insulator to a semiconductor.

Efforts in the 1980s and 1990s to incorporate impurities into the bulk crystal lattice of diamond via conventional doping procedures were unsuccessful in creating a room-temperature semiconducting material. This was mainly due to the high activation energy of acceptor and donor dopants, which are required to ignite mobile charge carriers in the semiconducting diamond bulk[2]. In the early 2000s, surface conductivity was discovered in hydrogen-terminated diamond[3], which led to the diamond surface transfer doping model[4]. This mechanism consists of electron exchange between a semiconductor (hydrogen-terminated diamond) and a surface dopant acceptor (aqueous molecules) with high electron affinity. A two-dimensional hole gas forms at the hydrogen-terminated diamond sub-surface and results in p-type surface conductivity at ambient conditions, with a sheet hole carrier concentration of around 10^{13} cm^{-2} and carrier mobility of several tens of $cm^2\ V^{-1}\ s^{-1}$.

Surface transfer doping of hydrogen-terminated diamond has led to

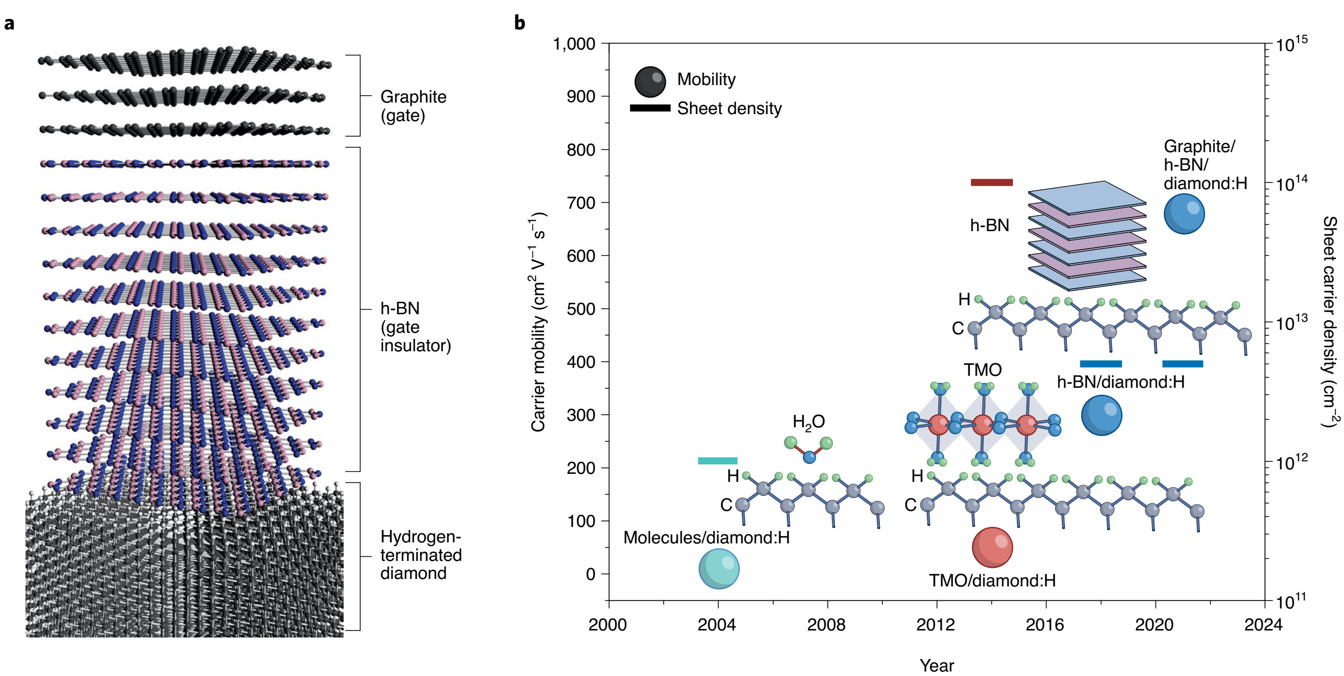

Fig. 1 | **Hydrogen-terminated diamond heterostuctures with enhanced carrier mobility. a**, Schematic diagram of the heterostructure consisting of graphite/hBN/hydrogen-terminated diamond. **b**, Benchmark comparison of various diamond surface conductivity performances at room temperature. Hole carrier mobilities and sheet carrier densities versus the year achieved. Also shown are schematics of the hydrogen-terminated diamond (diamond:H) heterostructures for different surface transfer doping methods. Molecular species interfaced with diamond:H (turquoise sphere), transition-metal oxides (TMO) interfaced with diamond:H (red sphere) and hBN interfaced with diamond:H (blue spheres), as reported by Takahide and colleagues[10,12]. Panel **a** reproduced with permission from ref. [10], Springer Nature Ltd.

the development of a number of diamond devices. However, these devices suffer from instability against temperature fluctuations and a loss in surface conductivity due to the volatile nature of the surface molecule acceptors[5]. Since 2013, transition-metal oxides have been employed as solid-state surface acceptors for hydrogen-terminated diamond[6–8]. This approach has improved the thermal robustness of the diamond surface conductivity (up to 450 °C), and led to record sheet hole carrier concentrations of 2×10^{14} cm^{-2}, but still with limited carrier mobilities of 10–100 cm^2 V^{-1} s^{-1} (ref. [9]). Writing in *Nature Electronics*, Yamaguchi Takahide and colleagues now report hydrogen-terminated diamond field-effect transistors (FETs) with ambient surface carrier mobilities as high as 680 cm^2 V^{-1} s^{-1} and sheet hole carrier densities of 5×10^{12} cm^{-2} (ref. [10]) (Fig. 1).

The researchers — who are based at the National Institute for Materials Science in Tsukuba and the University of Tsukuba — use laminated two-dimensional layers of monocrystalline hexagonal boron nitride (hBN) at the interface with the hydrogen-terminated diamond surface. The hBN/diamond heterojunction is shown to improve the quality of the two-dimensional hole gas channel by effectively reducing the number of surface charged impurities, which are the predominant cause of carrier scattering and is what has limited surface carrier mobility in previous hydrogen-terminated diamond FETs[11].

A FET based on this h-BN/diamond stack architecture was reported by Takahide and colleagues in 2018[12]. There, hBN was used as a gate dielectric and the device exhibited a carrier mobility of 300 cm^2 V^{-1} s^{-1} and a sheet hole carrier density of 5×10^{12} cm^{-2}. In their present work, the h-BN/diamond interface has been upgraded with an additional crystalline thin graphite layer on top, which acts as the gate electrode over the gate dielectric (hBN), further reducing disorder at the interface between the gate dielectric and the FET gate electrode.

These advances could allow diamond to be used in low-loss fast switching devices, high-frequency power amplifiers and other radio-frequency FET device applications. They also mean that diamond currently has the highest hole carrier (p-type) mobility of any wide-bandgap material (in comparison, GaN offers a carrier mobility of 30 cm^2 V^{-1} s^{-1} and SiC offers a carrier mobility of less than 17 cm^2 V^{-1} s^{-1}), as well as other superior intrinsic physical properties, including breakdown field and thermal conductivity. However, a number of challenges still need to be addressed in order to achieve the full potential of diamond electronics. The sheet hole carrier density needs to be improved to reduce the sheet resistance to below 1 kΩ sq^{-1} and the metal–diamond contact resistance should be lowered to the order of 0.1 Ω mm. Finally, room-temperature electron carrier (n-type) diamond conductivity needs to be achieved. ❐

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Competing interests
The author declares no competing interests.